The Science Behind a "Founder Score": Where to Start and How to Evaluate It

TL;DR

- **Individual founder traits are real but weak predictors:** the best-validated personality signals (Conscientiousness, Openness, low Neuroticism, need for achievement, generalized self-efficacy) correlate with entrepreneurial outcomes only modestly (individual correlations ~0.1–0.25; multivariate R ≈ .31 for performance). The single strongest *observable* predictor is prior founding success — Gompers et al. (2010) found a previously successful VC-backed entrepreneur has a 30% chance of succeeding again vs. 18% for a first-timer — which by definition is unavailable in the cold-start case. A defensible Founder Score should therefore be a **calibrated, wide-interval, probabilistic ranking signal**, not a precise point verdict.
- **Build it as a mechanical/actuarial model, not an AI oracle:** six decades of clinical-vs-statistical prediction research (Meehl; Grove et al. 2000) show simple mechanical combination of a few validated variables equals or beats expert intuition. For a cold-start founder, combine (a) validated psychometric constructs, (b) verifiable human-capital signals (domain expertise, founder-market fit, team composition), and (c) Bayesian shrinkage toward population base rates, wrapping every output in explicit uncertainty (credible/conformal intervals) and per-claim Trust Scores grounded in retrieved evidence.
- **Evaluate on ranking + calibration + process, not "accuracy":** because true outcomes take 5–10 years and the base rate of venture-scale success is tiny with power-law payoffs, evaluate with rank metrics (AUC, precision@k, NDCG) plus calibration (Brier, ECE, reliability curves), backtest on time-sliced historical cohorts with strict leakage control, measure inter-rater agreement vs human investors, and audit the LLM layer for citation faithfulness, test-retest consistency, and disparate impact.

Key Findings

1. **Personality predicts entrepreneurship weakly but consistently.** Meta-analyses (Zhao & Seibert 2006; Zhao, Seibert & Lumpkin 2010; Rauch & Frese 2007) converge: entrepreneurs score higher on Conscientiousness and Openness and lower on Neuroticism and Agreeableness, with small univariate effects but a moderate multivariate signal (R ≈ .37 for status, R ≈ .31 for performance). "Task-matched" traits (need for achievement, self-efficacy, innovativeness, stress tolerance, proactivity) outperform generic traits.
2. **The strongest predictor is unavailable for first-timers.** Prior entrepreneurial success is the most robust individual-level predictor. Gompers, Kovner, Lerner & Scharfstein (2010) found a previously successful VC-backed entrepreneur has a ~30% chance of success on the next venture, versus 18% for first-time entrepreneurs and 20% for those who previously failed.

This is the crux of the cold-start problem: the best feature is missing exactly when you need the score most.

3. **Trait inference from digital footprints is valid but ceiling-limited.** Language/like-based Big Five inference tops out around r ≈ .29–.40 (Azucar, Marengo & Settanni 2018 meta-analysis) — the "correlational upper limit" of behavior-to-personality. This compounds with the weak trait-to-success link, so a footprint→trait→success pipeline has two lossy stages.
4. **Statistically, founder prediction is genuinely hard.** Venture-scale startups fail at very high rates and venture returns follow a power law where a tiny fraction of deals drive most returns. Extreme class imbalance + heavy tails means no honest model can promise high precision on "who's the next unicorn"; the achievable goal is modest lift over base rate and good ranking/calibration.
5. **Cold-start has good analogues in credit scoring.** Thin-file credit scoring uses alternative data, hierarchical/empirical-Bayes shrinkage, and graph/network features — directly transferable patterns for scoring founders with no track record.
6. **VCs themselves bet on the team.** Gompers, Gornall, Kaplan & Strebulaev (2020) surveyed 885 VCs at 681 firms: the management/founding team was mentioned as an important factor by 95% of firms and as *the single most important* factor by 47% (vs. business model 83%, product 74%, market 68%, industry 31%). This validates a founder-centric score conceptually, even though it must be honest about limits.
7. **Industry ML VC is real but mostly a sourcing/triage tool.** EQT Motherbrain, SignalFire Beacon, Correlation Ventures, InReach, TRAC demonstrate value in surfacing and triaging deals — but even proponents concede AI "struggles when the data is sparse," and reported successes are sourcing wins, not validated success-probability calibration.
8. **Mechanical beats clinical.** Grove, Zald, Lebow, Snitz & Nelson (2000) meta-analysis: mechanical-prediction techniques were about 10% more accurate than clinical predictions on average, substantially outperformed them in 33%–47% of studies, and were substantially *worse* in only 6%–16% — strong justification for a structured, formulaic score over holistic partner intuition, with humans handling "broken-leg" exceptions.

Details

1. Psychology & psychometrics of entrepreneurial success

Big Five meta-analyses. Zhao & Seibert (2006, *Journal of Applied Psychology* 91(2):259–271; 19 studies, 24 samples, 133 effect sizes, 19,960 participants) found entrepreneurs score higher than managers on Conscientiousness and Openness, lower on Neuroticism and Agreeableness, with no Extraversion difference. Individual effect sizes were small but the multivariate relationship was moderate (R = .37). Zhao, Seibert & Lumpkin (2010, *Journal of Management* 36(2):381–404) extended this to intentions (multiple R = .36) and performance (multiple R = .31), with risk propensity predicting intentions but not performance. Rauch & Frese (2007, *European Journal of Work and Organizational Psychology* 16(4):353–385) showed traits *matched to the task* of running a business correlate more strongly with success (corrected r = .25) than unmatched traits (r ≈ .028); the strongest task-matched traits were need for achievement, generalized self-efficacy, innovativeness, stress tolerance, need for autonomy, and proactive personality.

Recent large-scale founder studies. McCarthy, Gong, Braesemann, Stephany, Rizoiu & Kern (2023, *Scientific Reports* 13:17200, DOI 10.1038/s41598-023-41980-y) analyzed a global sample (n = 21,187 startups; 32,732 founders clustered by Big Five facets inferred from Twitter/X). They found "we do not find one 'Founder-type' personality; instead, six different personality types appear" (the "FOALED" model: Fighters, Operators, Accomplishers, Leaders, Engineers, Developers). Key success-linked facets were Adventurousness (Openness — largest effect size), low Modesty (Agreeableness), and Activity Level (Extraversion). Critically, team composition mattered more than the individual: verbatim, "firms with three or more founders are more than twice as likely to succeed than solo-founded startups," and co-author Fabian Stephany (Oxford Internet Institute) noted that teams "with diverse combinations of types of founders have eight to ten times more chance of success than single founder organisations"; the study found founder personality roughly 5× more predictive than industry and 2× more than startup age. **Precision caveat:** their headline "82.5% accuracy" distinguishes *entrepreneurs from employees* from personality alone — it is NOT a startup-success prediction accuracy, though press coverage routinely conflates the two. Freiberg & Matz (2023, *PNAS* 120(19):e2215829120, DOI 10.1073/pnas.2215829120; 10,541 founder–startup dyads) found openness and agreeableness positively related to raising an initial round; neuroticism negatively related to all outcomes (resilience matters); conscientiousness positively related to early-stage investment but negatively to exit; extraversion not a significant predictor of any of four outcomes. nature nature

Grit & self-efficacy. Grit predicts success in some entrepreneurship samples (Mooradian et al. 2016, 281 Austrian entrepreneurs) but the evidence is contested — a large representative Spanish study found grit *modestly negatively* related to entrepreneurship involvement. Generalized self-efficacy and internal locus of control are among the more robust task-matched correlates (Rauch & Frese).

Team composition & founder-market fit. Evidence is mixed. Meta-analytic and empirical work finds mostly weak/insignificant links between founding-team skill heterogeneity and performance (Bell et al. 2011); some studies find educational-level heterogeneity helps time-to-unicorn, and STEM-founder + business-founder pairs show higher employment growth (Choi & Goldschlag). Solo founders sometimes outperform on survival in some contexts (Greenberg & Mollick). "Founder-market fit"/opportunity-founder fit is emphasized theoretically (Gruber) but is harder to measure psychometrically. Note that McCarthy et al.'s team-diversity multipliers and VCs' stated team emphasis both point the same direction: composition may carry more reliable signal than any single founder's traits.

Psychometric principles for the score. Any construct like "resilience" or "founder-market fit" must be treated as a latent variable measured with error. Design implications: (a) **reliability** — use multi-item/multi-source measurement and report test-retest and internal consistency; (b)

construct validity — anchor to validated instruments (IPIP/BFI facets) rather than ad-hoc LLM impressions; (c) **criterion validity** — the correlations above cap how much any trait can contribute; (d) **measurement error propagates** — a footprint→trait (r≈.3–.4) then trait→success (r≈.1–.25) pipeline must widen, not narrow, its intervals. Language-based assessments (Park et al. 2014, 66,732 Facebook users) show convergent validity ~r = .38 and 6-month test-retest ~r = .70, which sets a realistic reliability ceiling.

2. Statistics & predictive modeling under sparse/messy data

Base rates & power law. The oft-cited "~90% of startups fail" figure originates from the Startup Genome project and applies specifically to *venture-scale* startups; more granular US Bureau of Labor Statistics data on all private-sector businesses shows ~21.5% fail in year one, ~48.4% within five years, and ~65.1% within ten years. Venture returns are heavy-tailed — Correlation Ventures analyses of 21,000+ (2004–2013) and 27,000+ (2009–2018) investments, and Horsley Bridge/VenCap data, support a power-law fit (α ≈ 2.05). VenCap's analysis of 11,350 startups (1986–2018) found ~half lost money and just 121 (1.1%) returned the entire fund. Implication: with a ~1–2% "outlier" rate and extreme imbalance, precision on the extreme tail is intrinsically low; the honest deliverable is *lift over base rate* and *rank ordering*, plus wide uncertainty. Use the correct base rate for your defined population (all-comers vs. already-funded).

Cold-start / thin-file analogues. Credit bureaus score no-history applicants via alternative data (transactions, telecom, utilities), ensemble models, and graph/network features (relationships as predictors for thin-file borrowers). Empirical-Bayes shrinkage (Efron & Morris) estimates a population distribution of effects and shrinks sparse individual estimates toward it — directly applicable: a cold-start founder's score should be pulled toward sector/stage base rates until individuating evidence accumulates. FICO/VantageScore experience is sobering: among ~30–35M newly-scoreable thin-file consumers, only ~3% scored above 680, with an estimated ~30% bad rate — a caution that alternative data expands coverage but rarely produces confident "prime" scores for thin files.

Bayesian / hierarchical modeling & uncertainty. Use hierarchical priors (founder nested in sector × stage × geography), shrinkage toward population means, and report **credible intervals**. **Conformal prediction** provides distribution-free prediction sets with guaranteed marginal coverage under exchangeability — ideal for honest interval reporting on individual founders.

Calibration. A calibrated low-confidence score is more valuable than an overconfident point estimate. Measure with the **Brier score** (mean squared error of probabilities), **calibration/reliability curves**, and **Expected Calibration Error** (Naeini et al. 2015; Guo et al. 2017 showed modern ML tends to be overconfident). Caveat: the Brier score conflates calibration and the underlying base-rate difficulty, so report calibration curves/ECE alongside it, and apply post-hoc calibration (temperature/isotonic/Platt scaling).

Survivorship, selection & leakage. Crunchbase/PitchBook overrepresent survivors (failed startups delete profiles; funding data is incomplete), and training on such data inflates apparent success rates and biases features toward funding momentum. Label leakage is rampant: many published "success predictors" leak post-success information (e.g., later funding rounds) into features. Mitigations: strictly time-sliced features (as-of-date snapshots), out-of-time validation, PR-AUC + precision@k under realistic imbalance, inverse-prevalence weighting/SMOTE-NC only on training folds, and explicit modeling of the sampling/censoring process.

3. Existing data-driven VC & founder prediction

Academic ML. Many papers predict startup success from Crunchbase with headline AUCs of 0.77–0.97, but these are systematically inflated by leakage and survivorship (funding totals/rounds dominate feature importance — themselves consequences of success). A recent leakage-aware panel study (2010–2023 Crunchbase×USPTO) fit preprocessing on 2010–2019 and evaluated strictly out-of-time, using PR-AUC, Brier, and precision@k — the methodologically correct template. The recurring finding: financing recency/momentum, team size, and investor-network features dominate — which encode access and prior validation more than intrinsic founder quality.

Industry. EQT's Motherbrain (CNNs over ~50M companies, 50+ data sources) is credited with sourcing deals (AnyDesk; Small Giant Games→Zynga $700M; Wolt→DoorDash $8B) and lets EQT engage companies ~14 months before they fundraise; EQT reports it cut ~99% of deals early. SignalFire's Beacon tracks 600M+ data sources/half-a-trillion data points, scaling the firm to a $1B+ fund by 2025. Correlation Ventures scaled to ~4x typical deal count via model-driven diligence in days. TRAC.vc is fully algorithm-driven and cold-contacts founders (e.g., told Legacy it had "20% probability of becoming a unicorn within five years"). Uniform caveat: these are **sourcing/triage** systems whose value is speed and coverage; public evidence of *calibrated success probabilities* is scant, and TRAC concedes its approach requires founders to have "some existing digital footprint" — a direct cold-start limitation.

Fairness / bias. Algorithmic scoring of people replicates and can amplify network and demographic bias. Credit-scoring and hiring-algorithm literatures document **proxy discrimination** (neutral features standing in for protected attributes — e.g., "5 years of credit history" disadvantaging immigrants), **disparate impact** vs **disparate treatment**, and biases baked into historical training data. For founders, funding-momentum and network features are strong proxies for existing privilege (Sand Hill Road access, elite pedigree), risking a system that "predicts" success by rewarding those already advantaged. Mitigations: proxy audits, disparate-impact testing (80% rule), removing/regularizing proxy features, and keeping a human in the loop.

4. Model evaluation without ground truth / with delayed outcomes

- **Proxy labels & intermediate milestones:** since exit takes 5–10 years, use staged proxies (raised next round within 12 months, revenue/user milestones, patent-stock growth within 24 months, survival) while being explicit that these proxies themselves embed survivorship/funding bias.

- **Backtesting:** freeze features as-of historical dates, predict forward on cohorts with known outcomes, evaluate out-of-time. Report PR-AUC (not just ROC-AUC) given imbalance, precision@k (k = the number of founders you can realistically meet), and NDCG for ranked shortlists.
- **Ranking vs calibration:** for a "surface the top founders" use case, ranking metrics (AUC, precision@k, NDCG) are primary; but pair them with calibration so the probabilities attached to memos are honest.
- **Human agreement:** run inter-rater reliability studies between the score and experienced investors (and among investors themselves — expect modest agreement), and expert-agreement studies as an interim validity signal.
- **LLM pipeline evaluation:** LLM-as-judge is biased (position, verbosity, self-enhancement, and authority bias — favoring answers with citations *even if fabricated*) and non-deterministic. For an evidence-backed memo system: (a) enforce **retrieval grounding** and compute a **faithfulness/groundedness score** = fraction of claims supported by retrieved evidence (per-claim, the basis for Trust Scores); (b) use **self-consistency** (sample multiple reasoning paths; agreement signals reliability and flags hallucinations) for test-retest reliability; (c) measure hallucination/citation-faithfulness rates explicitly; (d) prefer binary/coarse judgments (LLMs are unreliable on fine-grained scoring scales) and pairwise comparisons where possible.

5. Practical synthesis — a scientifically defensible starting architecture

Design philosophy: a transparent, mechanical, Bayesian-shrunk score over a small set of validated constructs, wrapped in explicit uncertainty, with an LLM layer used for *evidence extraction and grounding* (not holistic judgment).

Constructs to measure (validated, task-matched):

1. Human capital / domain expertise & founder-market fit (from CV, LinkedIn, GitHub, publications) — most verifiable cold-start signal.
2. Prior founding/leadership experience and outcomes (even failed experience carries a small positive signal — Gompers found previously-failed founders at 20% vs. 18% for first-timers).
3. Task-matched dispositions where measurable: generalized self-efficacy, need for achievement, proactivity, stress tolerance/resilience, Openness/Adventurousness, Conscientiousness — measured via validated instruments where possible, and only cautiously inferred from language ($r \approx .3$–$.4$ ceiling).
4. Team composition & complementarity (size ⩾2–3, skill/role complementarity) — supported by McCarthy et al.'s finding that 3+ founder teams are >2x as likely to succeed and diverse-type teams 8–10x, and consistent with VCs' team emphasis. This may be your highest-signal cold-start feature.

How to combine: a simple, interpretable model — unit-weighted or logistic/linear with hierarchical shrinkage — over these constructs, calibrated post-hoc, outperforming and out-transparent-ing a black box, consistent with Meehl/Grove and Dawes' finding that even improper (unit-weighted) linear models rival optimally-weighted ones. Shrink cold-start scores toward sector×stage base rates; let the interval narrow only as verifiable evidence accumulates. Reserve human judgment for "broken-leg" exceptions the formula can't see.

Hackathon-scale evaluation plan (24–72h):

1. Assemble a time-sliced historical cohort (e.g., founders/startups as of year T with known outcomes by T+n) from Crunchbase-type data, deduping and acknowledging survivorship.
2. Enforce as-of-date features; no post-outcome leakage.
3. Report: ROC-AUC **and** PR-AUC, precision@k, NDCG for the shortlist; Brier score, ECE, and a reliability curve for calibration; base-rate lift.
4. LLM/memo layer: measure per-claim citation faithfulness (fraction of claims grounded in retrieved sources), hallucination rate on a spot-checked sample, and test-retest consistency across repeated runs (self-consistency).
5. Fairness spot-check: compare score distributions/precision across founder gender/geography/pedigree proxies; report disparate impact.
6. Communicate every score as a calibrated probability **with an interval** and a Trust Score per underlying claim.

Longer-term validation roadmap: prospective (forward-testing) cohorts with pre-registered proxy milestones; ongoing recalibration as outcomes mature; periodic drift monitoring; inter-rater studies vs investors; formal bias audits; and construct-validity studies linking inferred traits to validated psychometric instruments on a consented sub-sample.

Recommendations

1. **Frame the deliverable honestly:** ship a *calibrated probability with an explicit interval* and per-claim Trust Scores, not a single "Founder Score" number implying precision the science doesn't support. Benchmark to beat: base-rate lift and calibration, not raw accuracy.
2. **Start mechanical, stay interpretable:** implement a unit-weighted/logistic model over 4–6 validated constructs with Bayesian shrinkage toward sector/stage base rates. Add complexity only if it demonstrably improves *out-of-time* PR-AUC and calibration. *Threshold to change approach:* if a black-box model beats the linear one by a meaningful margin on out-of-time PR-AUC **and** calibration holds, consider it — otherwise keep the transparent model.
3. **Weight the team, not just the individual:** given that team composition shows more reliable signal (McCarthy et al.; VC survey) than any single trait, make founding-team size/complementarity a first-class feature.

4. **Use the LLM for grounding, not judging:** RAG over verifiable sources, per-claim faithfulness scoring, self-consistency sampling, coarse (not fine-grained) judgments, and explicit hallucination/citation audits. Treat any ungrounded claim as Trust Score 0.
5. **Engineer against leakage and survivorship from day one:** as-of-date snapshots, out-of-time splits, PR-AUC/precision@k, imbalance handling only on training folds.
6. **Audit for bias before shipping:** proxy-discrimination and disparate-impact checks; be explicit that funding-momentum/network features encode existing privilege and consider down-weighting them for the cold-start founder-quality signal.
7. **Keep a human in the loop:** structured expert review for exceptions and for the highest-stakes decisions, consistent with the mechanical-plus-broken-leg literature.

Caveats

- **What the literature says is NOT reliably predictable:** which specific early-stage founder becomes an outlier success. Individual trait→success correlations are small; the strongest predictor (prior success) is absent for first-timers; and power-law dynamics plus luck/timing (Gompers et al. found market timing explains much of serial-founder outperformance) cap achievable precision. Communicate this as irreducible uncertainty.
- **Two lossy inference stages:** footprint→trait (r≈.3–.4) and trait→success (r≈.1–.25) compound; do not present inferred-personality-based scores as high-confidence.
- **Data are biased:** Crunchbase/Twitter samples skew toward funded, tech, male, multi-founder, surviving companies (McCarthy et al. acknowledge this); models trained on them inherit those biases.
- **Headline numbers are often misread:** McCarthy et al.'s "82.5% accuracy" is entrepreneur-vs-employee classification, not success prediction; many Crunchbase AUCs of 0.9+ reflect leakage; and the "90% of startups fail" stat applies to venture-scale startups, not all businesses. Treat vendor and press claims skeptically.
- **LLM judgments are non-deterministic and biased** (position, verbosity, authority/citation bias); never rely on a single LLM pass for a score, and always verify citations.

Where to Start — Checklist

1. ☐ Define success proxies explicitly (next-round, survival, milestones) and acknowledge their biases; pick the correct base rate for your target population.
2. ☐ Build a time-sliced, leakage-controlled historical dataset; document survivorship gaps.
3. ☐ Pick 4–6 validated, task-matched constructs (domain expertise, founder-market fit, prior experience, self-efficacy/achievement/resilience, team complementarity).
4. ☐ Implement a simple linear/logistic model with hierarchical shrinkage toward sector×stage base rates.
5. ☐ Post-hoc calibrate; wrap outputs in conformal/credible intervals.

6. ☐ Add an LLM RAG layer with per-claim faithfulness (Trust) scoring and self-consistency.
7. ☐ Evaluate: PR-AUC, precision@k, NDCG + Brier, ECE, reliability curve + base-rate lift.
8. ☐ Audit citation faithfulness, hallucination rate, test-retest consistency, and disparate impact.
9. ☐ Ship calibrated probabilities with intervals and Trust Scores — never overconfident point verdicts.
10. ☐ Plan prospective validation, recalibration, and bias re-audits.